

Saba Dastpak

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EDUCATION

M.Sc. Mechatronics Engineering 2022 – 2025

Islamic Azad University, Karaj branch, Karaj, Iran

- GPA: (18.49 / 20)

B.Sc. electrical engineering 2017 – 2022

Imam Khomeini International University, Qazvin, Iran

- Full Scholarship
- GPA: (14.93 / 20)

RESEARCHES EXPERIENCE

Master's Thesis:

Improving Precision of Breast Cancer Detection in Mammogram Images 2024-2025

Islamic Azad University, Karaj branch, Karaj, Iran -- under the supervision of Dr. Haji Hosseini

- Applied image preprocessing techniques to improve classification and localization accuracy
- Employed and optimized YOLO parameters for tumor localization and classification

Research Assistant:

Microfluidic Channels & MEMS Technologies 2023

Imam Khomeini International University, Qazvin, Iran -- under the supervision of Dr. Hosseini

- Designed and fabricated microfluidic channels for categorizing cancerous versus non-cancerous cells based on impedance analysis, promoting advancements in biomedical diagnostics.
- Presented findings at the 2nd International Conference on New Researches and Technologies in Electrical Engineering (ICNRTEE).

PUBLICATIONS & JOURNALS

- Dastpak, S., & Hosseini, S. A. (2024). "Optimization of single-cell trapping using a transparent microfluidic chip." Presented at the 2nd International Conference on New Researches and Technologies in Electrical Engineering (ICNRTEE). Available at [Civilica](#)
- Dastpak, S., & Hajihosseini, P. "A YOLO-Based Model for Breast Cancer Detection in Mammograms" (Manuscript in preparation, based on Master's Thesis research)

SKILLS & TOOLS

Programming Languages:

- Python | MATLAB | JavaScript

Frontend & Web Development :

- React | Tailwind css | REST AP I Integration | i18next | Vite

AI, Machine Learning & Deep Learning :

- PyTorch | TensorFlow | Keras | Scikit-learn | CNNs | U-Net, YOLOv8 | Transfer Learning | Random Forests | SVM | KNN | Logistics Regression | Multi-Layer Perceptron (MLP)

Natural Language Processing (NLP):

- NLTK | Text Processing | Sentiment Analysis | Topic Modeling | Latent Semantic Analysis | Text Summarization | Spam Detection | Vector Space Models | Probabilistic and Markov Models

Computer Vision & Image Processing:

- OpenCV | CLAHE | Edge Detection | Morphological Operations | Segmentation, Thresholding | Feature Extraction | Object Detection

Model Optimization and Evaluation:

- Hyperparameter tuning | Data Augmentation | Regularization Techniques | AUC, F1, IoU

Development Tools:

- Git | Google Colab | Visual Studio | Pandas

Simulation and engineering Software

- COMSOL multiphysics | Revit | Pspice | Modelsim

Hardware and embedded systems

- arduino | AVR microcontrollers
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ENGLISH LANGUAGE PROFICIENCY

- **L2 English Teacher Training Course (TTC) Certificate**
Imam Khomeini International University, Qazvin, Iran | Aug 2021
- **TOEFL score:** 105 | Aug 2024

Reading

26

Listening

29

Speaking

27

Writing

23

ACADEMIC PROJECTS

- Utilizing Convolutional Neural Network (CNN) for classification, image segmentation, and object detection (YOLO)
- Comparison and implementation of various metric losses (Triplet-loss, cosine similarity, etc) for better representation in CNNs.
- Transfer Learning for Classification
- Comparative analysis of regression techniques for different datasets
- Exploring various regularization methods such as drop out, normalization, data augmentation, etc.
- Designing and implementing a chatbot capable of handling user orders and maintaining daily conversations using wit.ai service
- Developed a license plate recognition system using MATLAB on real-world vehicle images on MATLAB.
- Utilizing Multi Layer Perceptron (MLP) for regression and classification on various datasets
- Implemented sentiment analysis, spam detection, topic modeling, and text summarization using NLTK and Scikit-learn

ONLINE COURSERS AND CERTIFICATES

Coursera

- Coursera Introduction to Programming with MATLAB
- Convolutional Neural Networks in TensorFlow
- Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning
- Advanced Computer Vision with TensorFlow
- Object Localization with TensorFlow Deep Learning with PyTorch : Object Localization

Udemy

- [2026] Machine Learning: Natural Language Processing (V2)

WORKING EXPERIENCE

NorLayer | Self-Initiated Project

Jan 2026- present

- AI Hotel Chatbot Widget – Developed a multilingual, embeddable chatbot using React, Tailwind CSS, REST APIs, and i18next with dynamic configuration and theme support.

NovelBIM

Apr 2023- Aug 2023

Specialized Content Creator

- Developed technical content on electrical and mechanical systems, enhancing the platform's resources

Yadman Energy

Sep 2021 - Mar 2022

Smart Home Applications Intern

- Implemented smart technologies in building infrastructure, electrical panels and networking devices.

Azin Khodro Car Company

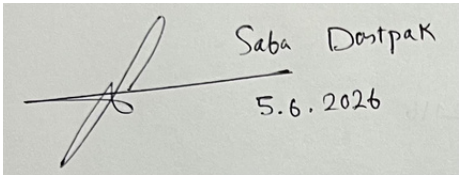
Jun 2020 - Sep 2020

Control & machine vision Intern

- Contributed to machine vision projects, emphasizing precision and system optimization.

EXTRACURRICULAR ACTIVITIES

- **Choir member**
Sorna Music Academy
- **English teacher Volunteer**



Saba Dostpak
5.6.2026

A handwritten signature in black ink, consisting of a stylized 'S' and 'D' connected together, followed by the name 'Saba Dostpak' and the date '5.6.2026' written in a simple, legible script.